PRETTY MARIA PURBA

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SUMMARY

Information Systems undergraduate with experience across software engineering, data analysis, system analysis, and user-centered design. Currently pursuing an independent final-year research internship at the National Research and Innovation Agency (BRIN), focusing on digital government, socio-economic data harmonization, multi-criteria decision analytics, and decision support systems for public planning. Previously worked as a Software Engineer Intern at PT Dimensi Kreasi Nusantara, contributing to the development of DOTS-CMS and DOTS-RIA by translating financial and regulatory business requirements into functional web and mobile features, dashboards, reporting workflows, and frontend–backend integrations using Next.js, React Native, TypeScript, REST APIs, and related technologies. Experienced in transforming complex business and data requirements into structured workflows, analytical models, dashboards, and digital products. Proficient in Python, JavaScript, Next.js, database concepts, API integration, statistical analysis, and UI/UX design. Demonstrated strong analytical, communication, and leadership skills through teaching assistant roles, research-based competitions, and cross-functional projects. Interested in data-driven information systems, digital government, analytics, software engineering, and AI-enabled solutions for real-world decision-making.

EDUCATION

Institut Teknologi Del – Sitoluama, Indonesia
Bachelor of Information Systems | 2023 – Present
Current GPA: 3.72/4.00

ACADEMIC PROJECTS

Bimbel Farmasi Web Application Development
Project Manager & Full-Stack Web Developer (Team-based) | Institut Teknologi Del – Web Application Programming and Testing Course | 2025

- Developed a full-stack web-based learning platform for pharmacy education (UKOM, CPNS/P3K, academic consultation).
- Implemented core features including authentication, program enrollment, transaction management, and content delivery using Laravel (PHP), MySQL, and REST API.
- Designed relational database structures and built responsive user interfaces (landing pages, dashboards, admin panels) to ensure efficient data processing and seamless user experience.
- Performed system testing and deployment to ensure reliability and usability across user roles.

LaporLink – Ticketing Web Application Development
Project Manager (Team-based) | Institut Teknologi Del – Information System Project Management Course | 2025

- Conducted requirement analysis and stakeholder discussions, defining functional & non-functional requirements for a web-based ticketing system to manage campus service reports efficiently.
- Created system design artifacts (BPMN, WBS, SRS) and designed user flows and wireframes to ensure clear system architecture and intuitive user experience.
- Applied Agile methodology (Scrum) to manage development, coordinating team execution and delivering core features such as ticket submission, status tracking, SLA monitoring, and community-based prioritization.

Pertanix Application System Design
Project Analyst & UI Designer (Team-based) | Institut Teknologi Del – System Analysis and Design Course | 2025

- Analyzed key challenges faced by farmers, including limited access to agricultural information, market prices, and difficulties in conducting efficient agricultural transactions.

- Designed an integrated digital system featuring three core modules: AI-based agricultural consultation, agricultural product sales, and agricultural product purchasing.
- Developed a comprehensive System Design Analysis document including use case diagrams, business process flows, and interface designs while also creating Figma-based UI prototypes and validating user interaction flows to ensure usability, efficiency, and seamless feature integration across the platform.

Evaluation and Redesign of Rumah Kreatif Toba Website
UI/UX Designer | Institut Teknologi Del – UI/UX Course | 2025
- Conducted heuristic evaluation, competitive analysis, and SUS measurement (scoring "Awful" at 31.75) for an MSME e-commerce platform.
- Identified 25+ high-severity usability issues and redesigned user flows and interactive UI prototypes in Figma, focusing on enhancing user experience and platform competitiveness.

Academic Record Management (T08 Transcript) - Java OOP Project
Software Developer (Pair Programming) | Institut Teknologi Del – Object-Oriented Programming Course | 2025
- Developed a Java console application for an Academic Simulator, culminating a series of weekly incremental assignments focusing on academic record management.
- Implemented complex logic to generate student transcripts, displaying only the most recent grade for repeatedly taken courses and ensuring chronological ordering.
- Applied advanced Object-Oriented Programming (OOP) concepts, including Nested Constructs, for enhanced code modularity and encapsulation.
- Demonstrated continuous feature development and refinement across multiple iterations of the same project case.

Business Process Analysis and Redesign for Student Permit System at IT Del
Business Process Analyst (Team-based) | Institut Teknologi Del – Business Process Management Course | 2024
- Identified and analyzed inefficiencies in campus permit processes via stakeholder elicitation.
- Redesigned workflows using Task Elimination, Parallelism, and Automation, documented with BPMN and process simulations, resulting in a structured SRS-equivalent report.

WORK EXPERIENCE

Research Intern
Badan Riset dan Inovasi Nasional – Digital Government Research Group | September 2026 - Present

Software Engineer Intern
PT Dimensi Kreasi Nusantara | February 2026 – July 2026
- Developed responsive web and mobile interfaces for financial technology products using Next.js, React Native, TypeScript, and REST APIs.
- Contributed to DOTS-CMS and DOTS-RIA modules, translating complex financial workflows and regulatory requirements into structured user interfaces and system functionality.
- Participated in requirement analysis, functional testing, UAT preparation, technical documentation, and iterative product development using Agile/Scrum practices.

Teaching Assistant – Algorithms and Data Structure
Institut Teknologi Del | August 2025 – December 2025
- Assisted in preparing practical materials and example problems related to data structures (arrays, linked lists, stacks, queues, trees, graphs) and fundamental algorithms (sorting, searching).
- Contributed to improving students' comprehension and problem-solving skills.

Teaching Assistant – Probability and Statistic
Institut Teknologi Del | August 2025 – December 2025
- Provided academic assistance for foundational statistics topics such as measures of central tendency, dispersion, probability, and normal/binomial distributions.
- Reviewed assignments and provided constructive feedback to improve students' analytical and computational skills.
- Supported students in debugging code and enhancing their understanding of algorithm complexity analysis.

Teaching Assistant – Probability and Statistic
Institut Teknologi Del | January 2025 – May 2025

- Prepared supplementary materials, exercises, and explanations on topics including probability theory, probability distributions, parameter estimation, and hypothesis testing. Provided academic assistance for foundational statistics topics such as measures of central tendency, dispersion, probability, and normal/binomial distributions.
- Assisted students in applying statistical tools (Excel) for basic data analysis tasks.
- Facilitated discussion sessions to strengthen conceptual understanding and interpretation of statistical results.

Teaching Assistant – English 2
Institut Teknologi Del | January 2025 – May 2025

- Supported teaching activities focused on academic writing, presentation skills, and reading comprehension.
- Guided students in improving communication skills through presentation practice, discussions, and structured writing exercises.
- Facilitated language practice sessions to enhance grammar accuracy, vocabulary usage, and overall confidence.

Teaching Assistant – English 1
Institut Teknologi Del | August 2024 – December 2024

- Led conversation and discussion exercises to help students build speaking fluency.
- Assisted in delivering foundational English materials such as grammar essentials, sentence structure, and reading comprehension.
- Reviewed written assignments and provided feedback to improve students' accuracy and clarity in English.

ORGANIZATIONAL EXPERIENCE

Del Debate Championship – Institut Teknologi Del
Chief Organizer | March 2026 – May 2026

- Led the end-to-end execution of Del Debate Championship 2026, a campus-level debate competition following NUDC standards, involving 30 participants and multiple stakeholders.
- Coordinated cross-functional committees (consumption, equipment, administration, documentation, and MC) to ensure smooth event operations and timely delivery.
- Managed event planning, scheduling, and logistics, including participant registration, venue preparation, and judge coordination across multiple competition days.

Del English Club (DEC) - Institut Teknologi Del
Leader | August 2025 – Present

- Coordinated club activities and supervised weekly English practice sessions to improve members' speaking, writing, and presentation skills.
- Designed learning programs, workshops, and collaborative activities aimed at enhancing students' English proficiency and confidence.
- Led internal meetings, delegated tasks to team members, and ensured effective execution of club events and competitions.

Student Executive Board (BEM) - Institut Teknologi Del
Treasurer of Department of Facilities and Infrastructure | December 2024 – Present

- Actively involved in managing and evaluating campus facilities to support an optimal student environment.
- Assisted in coordinating infrastructure development initiatives and monitored maintenance-related activities.

Information Systems Student Association (HIMSI) - Institut Teknologi Del
Member | January 2025 – Present

- Participated in planning and executing educational programs for Information Systems students.
- Involved in technology socialization activities (Makey Makey, Edison Robot) for elementary school students, introducing interactive tech.

TRY OUT PRE-UTBK - Institut Teknologi Del
Head of Public Engagement Division | March 2025 – April 2025

- Led the division responsible for external communication, participant coordination, and outreach to prospective students.
- Oversaw the dissemination of event information across social media and offline channels, increasing participant engagement.
- Coordinated with schools, committees, and volunteers to ensure smooth registration and event execution.

ACHIEVEMENTS

3rd Place – Del AI Hackathon 2026
Institut Teknologi Del

- Developed TIUR, a tourism intelligence solution that transformed 22,144+ traveler reviews into structured service-level insights for tourism decision-making.
- Applied LLM-assisted Aspect-Based Sentiment Analysis, rating-sentiment conflict analysis, Alignment Index, Importance-Performance Analysis, and evidence-based retrieval to identify priority tourism service issues.
- Contributed to analytical workflow design, system conceptualization, dashboard development, and presentation of actionable insights for tourism stakeholders.

Silver Medal – International Future Leaders Competition (IFLC), 2026
Sentosa Foundation & Universitas Gadjah Mada

- Contributed to the development of a research-driven system titled "From Invisible to Investable: Empowering MSMEs Through WhatsApp-First AI Finance", focusing on AI-enabled financial system design for MSMEs.
- Involved in system conceptualization and solution architecture, including AI-based financial workflow, MSME digitization logic, and platform design thinking.

Top 5 Finalist – Business Plan Competition (2026)
Universitas Atma Jaya Yogyakarta

- Achieved 4th Place Overall among finalist teams in a competitive business plan evaluation.
- Contributed to the design and system thinking of REWEAVE – Circular Textile Supply Chain Platform, focusing on marketplace architecture, inventory digitization, and workflow design.
- Involved in defining core system components including digital inventory structure, supplier-buyer workflow, trust layer (escrow & grading system), and AI-based material matching concept (filter-based MVP design).

Silver Medal – Lombok Essay Competition 6 (LEC 6), 2026
Ruang Inovasi Sains dan Karya Ilmiah (RISKI) & Fakultas Pertanian Universitas Muhammadiyah Mataram

- Recognized for developing a data-driven solution addressing educational inequality through geospatial mapping and accessibility analysis.
- Demonstrated strong capability in research, innovation design, and problem-solving in a national-level essay competition.

Best Adjudicator – Del Debate Championship (2025)
Institut Teknologi Del

- Recognized as the Best Adjudicator for outstanding analytical judgment, decision-making accuracy, and clarity in delivering debate evaluations.
- Contributed to event preparation and coordination, ensuring smooth competition flow and fair assessment standards for participants.
- Demonstrated strong critical thinking, communication, and teamwork skills in managing debate sessions and collaborating with the organizing committee.

CORE COMPETENCIES

- **Languages:** Indonesian (Fluent), English (Intermediate)
- **Technical Skills:**
  Programming: C, Java, JavaScript, Python
  Web & Mobile: Next.js, React Native
  Software Engineering: Object-Oriented Programming (Java), Data Structures & Algorithms, API Integration, Microservices Fundamentals
  Data & Analysis: Statistical Analysis (Excel), Database Concepts
- **Tools:** Figma, Draw.io, Trello/Asana, Git, Microsoft Office (Word, Excel, PowerPoint), Postman
- **Soft Skills:** Critical Thinking & Analytical Reasoning, Problem-Solving, Communication & Presentation, Teaching & Mentoring, Teamwork & Collaboration, Leadership & Coordination, Event Management, Adaptability, Time Management, Decision-Making